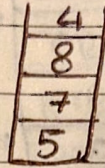


Stacks and Queues

Stacks

S = [5, 7, 8, 4] $\Rightarrow$



^{LIFO}
Last In First Out

* Append (x):

$\rightarrow$ if dynamic array

* $O(1)$

$\rightarrow$ used most of the time

$\rightarrow$ if doubly linked list

$O(1)$

* pop:

$O(1)$

* Is Empty: $O(1)$

if S:

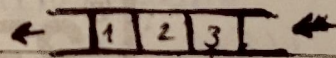
* peek: $O(1)$

$\hookrightarrow$ what's on top of the stack $\Rightarrow S[-1]$

Queues

FIFO $\rightarrow$ First In First Out

[1, 2, 3]



* Enqueue (x) $\rightarrow$ if dynamic array * $O(1)$

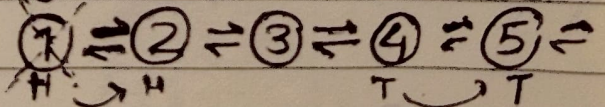
$\hookrightarrow$ just like append

$\rightarrow$ if doubly linked list $O(1)$

* Dequeue

if dynamic array [1, 2, 3, 4] $\rightarrow$ [2, 3, 4, x]

$\hookrightarrow$ if doubly linked list



python

$\hookrightarrow$ from collections import deque

double ended queue

(insert/delete from both sides)